Notes: Chen, He, and Liu (JFE, 2020)

The Financing of Local Government in China: Stimulus Loan Wanes and Shadow Banking Waxes

Contents

1 What the paper is doing

Big question. I want to understand why China's *shadow banking* (trust/entrusted loans, wealth management products) and *bond financing for local governments* expanded so rapidly *after 2012*, even though the original policy shock (the 2009 stimulus plan) happened years earlier.

Core mechanism (the "stimulus loan hangover" idea).

- China's 2009 stimulus triggered a very large, uneven (across provinces) expansion of *bank loans*, especially loans routed to local-government financing vehicles (LGFVs).
- These stimulus-era LGFV loans created future *rollover* and *continuing-investment* needs.
- Around 2012–2015, as the traditional banking channel tightened, the financing pressure shifted toward *municipal corporate bonds (MCBs)* and broader shadow-banking channels.

Empirical design. The paper builds a cross-province measure of the 2009 stimulus lending surge and tests whether it predicts future MCB issuance and shadow-banking activity, especially in 2012–2015.

Main contribution. The paper links (i) the 2009 fiscal/credit stimulus to (ii) the later rise of bond-financed local government debt and shadow banking, and interprets the sequence as a delayed financing reallocation driven by loan maturity/rollover constraints and regulatory tightening.

2 Institutional background and timeline

2.1 Objects in the paper

- **LGFVs**: off-budget vehicles that borrow and invest on behalf of local governments.
- **Stimulus-era bank loans (2009)**: bank credit expansion encouraged by policy, partly flowing into LGFVs.
- **MCBs**: "municipal corporate bonds" issued by LGFVs; in practice, these look like quasi-municipal debt with implicit guarantees.
- **Shadow banking**: entrusted loans, trust loans, wealth management products (WMP), and related off-balance-sheet channels.

2.2 Key timing emphasized by the paper

- **2009**: huge spike in new bank lending (stimulus year).
- **2010–2011**: policies attempt to rein in LGFV borrowing; MCB issuance is suppressed.
- **2012–2015**: MCB issuance (and shadow banking) grows rapidly — interpreted as the delayed response to refinancing needs created by 2009 loans.

- **2015**: debt-swap program begins (munibond swaps), reflecting recognition of refinancing stress and implicit guarantees.

3 Data construction and measurement (the paper's key variables)

The empirical analysis is built around two constructed objects: a province-level measure of the 2009 stimulus lending shock, and a measure of abnormal MCB issuance relative to the pre-stimulus baseline.

3.1 (Model/Definition 1) 2009 stimulus bank-loan measure

For each province i, define the 2009 stimulus bank loan (scaled by 2009 GDP) as the deviation from its pre-2009 average:[1]

$$\frac{\mathrm{BL}_{i,2009}^{\mathrm{stimulus}}}{\mathrm{GDP}_{i,2009}} \equiv \frac{\mathrm{BL}_{i,2009}}{\mathrm{GDP}_{i,2009}} - \frac{1}{5}\sum_{\tau=2004}^{2008} \frac{\mathrm{BL}_{i,\tau}}{\mathrm{GDP}_{i,\tau}}. \tag{1}$$

Interpretation. This normalizes the stimulus intensity by removing a province's “steady-state” credit intensity. So a high value means an unusually aggressive local credit expansion in 2009.

3.2 (Model/Definition 2) Abnormal MCB issuance

For each province i and year t, define abnormal MCB issuance (scaled by same-year GDP) as the deviation from the 2004–2008 mean:[2]

$$\frac{\mathrm{MCB}_{i,t}^{\mathrm{abnormal}}}{\mathrm{GDP}_{i,t}} \equiv \frac{\mathrm{MCB}_{i,t}}{\mathrm{GDP}_{i,t}} - \frac{1}{5}\sum_{\tau=2004}^{2008} \frac{\mathrm{MCB}_{i,\tau}}{\mathrm{GDP}_{i,\tau}}, \qquad t \in \{2012, 2013, 2014, 2015\}. \tag{2}$$

Interpretation. This removes the pre-stimulus baseline level of MCB issuance.

3.3 MCB purpose decomposition

MCB issuance is split into three categories:

- $\mathrm{MCB}^{\mathrm{repay}}$: issuance used to repay bank loans (rollover motive),
- $\mathrm{MCB}^{\mathrm{inv}}$: issuance used for investment (continuation motive),
- $\mathrm{MCB}^{\mathrm{other}}$: other stated purposes.

Interpretation. This is crucial: the paper does not just show “more bonds” — it tries to map bond issuance to economic motives consistent with the hangover mechanism.

[1] This is Eq. (1) in the paper.
[2] This is Eq. (2) in the paper.

4 Empirical models (all main equations) and what each coefficient means

4.1 (Model 3) Year-by-year cross-sectional regression (Table 3)

The simplest test is year-by-year cross-sectional prediction:[3]

$$\frac{\mathrm{MCB}_{i,t}^{\mathrm{abnormal}}}{\mathrm{GDP}_{i,t}} = a_t + b_t \frac{\mathrm{BL}_{i,2009}^{\mathrm{stimulus}}}{\mathrm{GDP}_{i,2009}} + u_{i,t}. \tag{3}$$

Interpretation.

- $b_t > 0$ in 2012–2015 means: provinces with larger 2009 credit surges issue more abnormal MCBs several years later.
- The "hangover" prediction is that b_t should be small before the refinancing pressure hits, and becomes large when rollover needs bind (post-2012).

4.2 (Model 4) Panel event-study style regression (Table 4; Fig. 5)

The main specification uses a panel with province and year fixed effects, and interacts the 2009 shock with year dummies:[4]

$$\frac{\mathrm{MCB}_{i,t}}{\mathrm{GDP}_{i,2009}} = \sum_{\tau \in T \setminus \{2008\}} \beta_\tau \, \mathbf{1}_\tau(t) \, \frac{\mathrm{BL}_{i,2009}^{\mathrm{stimulus}}}{\mathrm{GDP}_{i,2009}} + \alpha_i + \alpha_t \tag{1}$$

$$+ \sum_{\tau \in T \setminus \{2008\}} \gamma_\tau' \, \mathbf{1}_\tau(t) \, X_{i,0704\text{–}0803} + u_{i,t}, \tag{4}$$

where $T = \{2004, \ldots, 2015\}$ and the year 2008 is the omitted benchmark.

Why scale by $\mathrm{GDP}_{i,2009}$**?** Fixing the denominator at 2009 GDP prevents the dependent variable from mechanically moving with future GDP (which could itself be affected by stimulus loans).

Interpretation of β_τ**.** β_τ is the differential MCB issuance in year τ associated with a one-unit larger 2009 stimulus-loan shock. The hangover mechanism predicts:

- pre-2009: $\beta_\tau \approx 0$ (no pre-trends),
- 2009–2011: $\beta_\tau > 0$ but modest (banks are still lending),
- 2012–2015: β_τ becomes large and significant (rollover + tightening shifts financing to bonds/shadow banking).

[3] This is Eq. (3) in the paper.
[4] This is Eq. (4) in the paper.

Controls $X_{i,0704\text{–}0803}$**.** These are pre-stimulus province characteristics measured around 2007Q4–2008Q3, including fiscal deficit/GDP, fixed asset investment/GDP, GDP per capita, GDP growth, and a Big-Four bank branch-share proxy for the sophistication/dominance of large banks.

4.3 Identification and endogeneity discussion (instrumental variable approach)

A key concern is that provinces with high stimulus lending might differ systematically (e.g., in development, infrastructure needs, political priorities) and those differences might also drive later bond issuance.

4.3.1 Instrument: governor tenure dummy $\mathrm{LateTerm}_{i,2009}$

The paper uses a political-economy instrument based on the governor's term:

- $\mathrm{LateTerm}_{i,2009} = 1$ if the provincial governor in 2009 had served *more than two years* in office by 2009, and 0 otherwise.
- Motivation: promotion incentives and policy execution can vary over the official term.

First stage (conceptual). In the first stage, the stimulus-loan measure is predicted by $\mathrm{LateTerm}_{i,2009}$ (plus controls):

$$\frac{\mathrm{BL}^{\mathrm{stimulus}}_{i,2009}}{\mathrm{GDP}_{i,2009}} = \pi_0 + \pi_1\,\mathrm{LateTerm}_{i,2009} + \Pi' X_{i,0704\text{–}0803} + \nu_i. \tag{2}$$

Second stage. Replace $\mathrm{BL}^{\mathrm{stimulus}}_{i,2009}/\mathrm{GDP}_{i,2009}$ in Eq. (4) with its fitted value.

Exclusion restriction (as I understand it). Conditional on pre-2009 fundamentals and fixed effects, governor tenure in 2009 should affect post-2012 MCB issuance primarily through its effect on 2009 stimulus lending intensity (the hangover channel), not through other persistent province-specific shocks.

Caveat: weak instrument. The paper emphasizes that the first-stage F-statistics are relatively low, so IV magnitudes should be interpreted cautiously.

4.4 (Model 5) Heterogeneity: the role of long-maturity policy-bank loans (Table 6)

The paper predicts weaker rollover pressure when 2009 lending relied more on long-maturity policy-bank loans (especially from China Development Bank). They define $\mathrm{CDB}^{\mathrm{low}}_{i,2009}$ as an indicator for provinces with below-median 2009 CDB-loan fraction.

The heterogeneity specification is:[5]

$$\frac{\mathrm{MCB}_{i,t}}{\mathrm{GDP}_{i,2009}} = \sum_{\tau \in T\setminus\{2008\}} \beta_\tau^{\mathrm{inter}} \, \mathbf{1}_\tau(t) \frac{\mathrm{BL}_{i,2009}^{\mathrm{stimulus}}}{\mathrm{GDP}_{i,2009}} \, \mathrm{CDB}_{i,2009}^{\mathrm{low}} \tag{3}$$

$$+ \sum_{s \in \{1,2,3\}} \beta_s^{\mathrm{BL}} \, \mathbf{1}_s(t) \frac{\mathrm{BL}_{i,2009}^{\mathrm{stimulus}}}{\mathrm{GDP}_{i,2009}} + \sum_{\tau \in T\setminus\{2008\}} \beta_\tau^{\mathrm{CDB}} \, \mathbf{1}_\tau(t) \mathrm{CDB}_{i,2009}^{\mathrm{low}} \tag{4}$$

$$+ \alpha_i + \alpha_t + \sum_{\tau \in T\setminus\{2008\}} \gamma_\tau' \, \mathbf{1}_\tau(t) \, X_{i,0704\text{–}0803} + u_{i,t}. \tag{5}$$

Here $\mathbf{1}_s(t)$ are subperiod indicators (2004–2007, 2009–2011, 2012–2015).

Interpretation.

- $\beta_\tau^{\mathrm{inter}} > 0$ post-2012 means: the hangover effect is stronger in provinces with *low* CDB-loan fractions, consistent with more short-maturity commercial-bank lending creating more refinancing pressure.
- This supports the maturity/rollover channel rather than a generic "stimulus demand" story.

5 Reading the figures and tables

In the reading, the paper is organized as: (i) stylized facts → (ii) shock construction → (iii) dynamic treatment effects → (iv) mechanism/heterogeneity → (v) shadow-banking link → (vi) broader context.

5.1 Fig. 1: aggregate bank-loan surge in 2009

What I see. New bank loans jump sharply in 2009 relative to GDP and relative to 2004 GDP. The 2009 spike is visually distinct from 2004–2008 and from post-2010 levels.

Why it matters. This provides the aggregate time-series motivation: there is a clear policy-linked credit expansion to explain.

5.2 Table 1 and Fig. 2: local government debt composition shifts toward non-bank channels

Table 1. Using National Audit Office (NAO) reports, total local government debt rises substantially from 2010/12/31 to 2013/6/30, and the composition shifts:

- bank-loan share falls (from about 79% to about 57%),
- bonds increase,

[5] This is Eq. (5) in the paper.

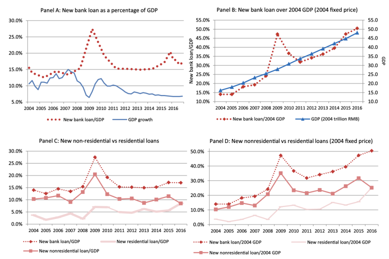


Fig. 1. New bank loan growth in China, 2004–2016. Panel A plots annual new bank loans over GDP and quarterly GDP growth; Panel B plots new bank loans over 2004 GDP (left scale) and GDP in trillion RMB in 2004 fixed price (right scale); Panel C plots new bank loans, new nonresidential bank loans, and new residential bank loans, all over GDP; and Panel D plots new bank loans, new nonresidential bank loans, and new residential bank loans, all over 2004 GDP and in 2004 fixed price. Numbers in fixed 2004 price are converted using GDP deflator. Data source: People's Bank of China (bank loans) and China National Bureau of Statistics (GDP and GDP deflator).

Table 1
Local government debts from NAO reports.
This table reports sources of local government debt from two reports issued by the National Auditing Office of China. Panels A and B report the local government debt outstanding by category in value (RMB billion) and percentage. Seven forms of local government debt include bank loan, fiscal on-lending, bond, other entity and individual borrowing, build and transfer, trust, and others. Two reports are as of December 31, 2010 and June 30, 2013.

Panel A: RMB billion			
	2010/12/31	2013/6/30	Change
Bank loan	8468	10,119	*1651*
Fiscal on-lending	448	303	*−145*
Bond	757	1846	*1089*
Other entity and individual borrowing	1045	839	*−206*
Build and transfer	0	1476	*1476*
Trust	0	1425	*1425*
Others	0	1882	*1882*
Total	10,717	17,891	*7174*

Panel B: Percentage			
	2010/12/31 (%)	2013/6/30 (%)	Change (%)
Bank loan	79.0	56.6	*−22.4*
Fiscal on-lending	4.2	1.7	*−2.5*
Bond	7.1	10.3	*3.2*
Other entity and individual borrowing	9.8	4.7	*−5.1*
Build and transfer	0.0	8.3	*8.3*
Trust	0.0	8.0	*8.0*
Others	0.0	10.5	*10.5*

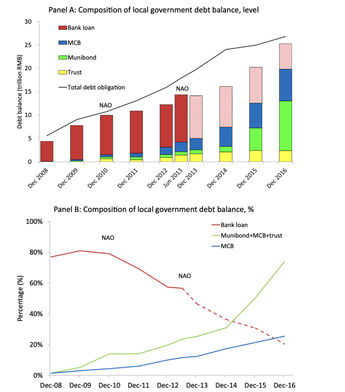


Fig. 2. Local government debt composition, 2008–2016. Panel A plots the composition of local government debt balance in trillion RMB, and Panel B plots the percentage of local government debt balance by composition. Four forms of local government liability include bank loan, municipal corporate bond, municipal bond, and trust. After 2013, LGFV bank loans are indicated by the shaded area/dashed line, as our estimate is based on China Construction Bank only and hence is less accurate. The data construction details are in the Online Appendix B.

- trust and other nonbank categories become sizable.

Fig. 2. Over 2008–2016, the share of bank loans in local government debt declines while MCBs and other nonbank debts rise.

Why it matters. This is the macro "reallocation" fact the paper wants to explain: the system transitions from bank-loan financing to market/shadow financing.

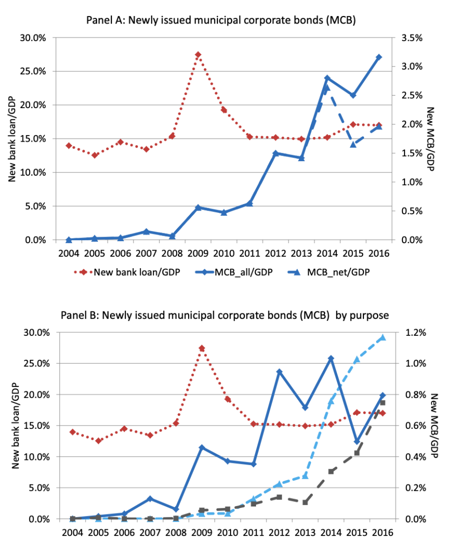


Fig. 3. Municipal corporate bond issuance, 2004–2016. Panel A plots the total MCB issuance over GDP, the net MCB issuance over GDP, and new bank loans over GDP. Panel B plots the total MCB issuance over GDP by usage, including repayment of bank loans, financing of an investment, and other purposes (including replenishing working capital, financing for other entities through entrusted loan structure, repayment of trust loans or other financial institutional borrowings, and undisclosed purpose). New bank loans over GDP are plotted against the left vertical axis, and MCB issuance over GDP is plotted against the right vertical axis. The annual MCB-issuance data are aggregated from individual municipal corporate bonds downloaded from WIND.

Table 2
Summary statistics.
This table reports the summary statistics of key variables for provincial MCB issuance and economic conditions. Panel A reports the summary statistics of all variables over the full sample. Panels B and C report the summary statistics of MCB issuance over the 2004–2008 and the 2009–2015 subperiods. Dependent variables include MCB over GDP, MCB for repayment of bank loans over GDP, MCB for investment over GDP, MCB for other purpose over GDP, fixed asset investment over GDP, and GDP per capita, all of which are scaled by 2009 GDP. The main explanatory variable is stimulus bank loan, defined as 2009 bank loans over GDP minus its average value over the past five years. Control variables include fiscal deficit over GDP, fixed-asset investment over GDP, GDP growth, GDP per capita (in RMB thousand), and the Big Four branch share, the former four of which are measured over the one-year window of 2007Q4–2008Q3, and the last one is measured as of 2008Q3. The sample period for dependent variables are from 2004 to 2015, except for the entrusted loan (EL)/GDP with the sample period of 2013–2015.

Panel A: Summary statistics of full sample								
	Obs	Mean	SD	Min	P25	Median	P75	Max
$MCB_{i,t}/GDP_{i,2009}$	360	0.016	0.027	0.000	0.000	0.003	0.020	0.162
$MCB^{repay}_{i,t}/GDP_{i,2009}$	360	0.004	0.010	0.000	0.000	0.000	0.003	0.079
$MCB^{inv}_{i,t}/GDP_{i,2009}$	360	0.007	0.010	0.000	0.000	0.002	0.009	0.054
$MCB^{other}_{i,t}/GDP_{i,2009}$	360	0.002	0.004	0.000	0.000	0.000	0.002	0.032
$BL/GDP^{stimulus}_{i,2009}$	30	0.164	0.050	0.083	0.129	0.158	0.190	0.270
$FD/GDP_{i,07Q4-08Q3}$	30	0.086	0.064	−0.002	0.026	0.087	0.105	0.274
$FAI/GDP_{i,07Q4-08Q3}$	30	0.475	0.121	0.236	0.400	0.452	0.557	0.710
$\Delta GDP_{i,07Q4-08Q3}$	30	0.126	0.024	0.086	0.104	0.128	0.136	0.187
$GDP^{capita}_{i,07Q4-08Q3}$	30	25.479	14.649	9.087	16.737	19.580	31.645	65.803
$BigFour_{i,2008Q3}$	30	0.381	0.081	0.237	0.331	0.358	0.439	0.570
$FAI_{i,t}/GDP_{i,2009}$	360	0.847	0.619	0.149	0.330	0.662	1.215	2.969
$GDP^{capita}_{i,t}/GDP^{capita}_{i,2009}$	360	1.184	0.527	0.321	0.724	1.050	1.631	2.721
$EL/GDP_{i,t}$	90	0.033	0.026	−0.018	0.017	0.027	0.041	0.142

Panel B: Summary statistics of MCB issuance for the 2004–2008 subperiod								
	Obs	Mean	SD	Min	P25	Median	P75	Max
$MCB_{i,t}/GDP_{i,2009}$	150	0.000	0.001	0.000	0.000	0.000	0.000	0.009
$MCB^{repay}_{i,t}/GDP_{i,2009}$	150	0.000	0.000	0.000	0.000	0.000	0.000	0.001
$MCB^{inv}_{i,t}/GDP_{i,2009}$	150	0.000	0.001	0.000	0.000	0.000	0.000	0.009
$MCB^{other}_{i,t}/GDP_{i,2009}$	150	0.000	0.000	0.000	0.000	0.000	0.000	0.001

Panel C: Summary statistics of MCB issuance for the 2009–2015 subperiod								
	Obs	Mean	SD	Min	P25	Median	P75	Max
$MCB_{i,t}/GDP_{i,2009}$	210	0.027	0.032	0.000	0.006	0.016	0.036	0.162
$MCB^{repay}_{i,t}/GDP_{i,2009}$	210	0.007	0.013	0.000	0.000	0.002	0.009	0.079
$MCB^{inv}_{i,t}/GDP_{i,2009}$	210	0.011	0.011	0.000	0.003	0.008	0.015	0.054
$MCB^{other}_{i,t}/GDP_{i,2009}$	210	0.003	0.005	0.000	0.001	0.001	0.004	0.032

5.3 Fig. 3: municipal corporate bond issuance over time and by purpose

What I see. MCB issuance is tiny pre-2009, rises around the stimulus period, is suppressed in 2010–2011, then accelerates strongly after 2012. The purpose split suggests that a large component is tied to bank-loan repayment/rollover.

Why it matters. This is exactly the delayed-timing fact: the big boom is not immediate in 2009, but later.

5.4 Table 2: summary statistics

What I take away.

- Province-level stimulus-loan deviations are economically meaningful (nontrivial dispersion).
- MCB issuance is near-zero in 2004–2008, and materially positive in 2009–2015.
- The paper has both province-level (N=30) and city-level panels (large N) for robustness.

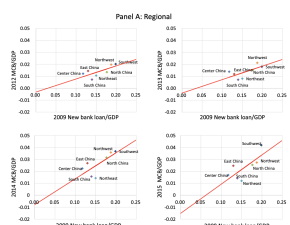
5.5 Table 3 and Fig. 4: preliminary cross-sectional evidence

Model link. These correspond to Eq. (3).

What Table 3 shows. The coefficient b_t is positive for 2012–2015 across three aggregation levels (region, province, city). In other words, high-2009-stimulus areas later issue more abnormal MCBs.

Table 3
The effects of 2009 stimulus bank loan on future municipal corporate bond issuance, year-by-year regressions.
This table reports the year-by-year regressions of 2012–2015 MCB issuance on 2009 bank loan. The dependent variable is the abnormal MCB issuance scaled by GDP in years 2012–2015 compared to the average value between 2004 and 2008. Annual MCB issuance at the regional/provincial/city level is aggregated over individual MCB bonds. The explanatory variable is the stimulus bank loan scaled by GDP. Panels A, B, and C report the cross-regional, the cross-provincial, and the cross-city results, respectively. Data on bank loans are obtained from the PBoC, and data on MCBs are obtained from WIND. Constants are not reported. Heteroskedasticity consistent standard errors are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	(1) MCB/GDP_{2012}	(2) MCB/GDP_{2013}	(3) MCB/GDP_{2014}	(4) MCB/GDP_{2015}
Panel A: Regional regressions				
$BL/GDP^{stimulus}_{2009}$	0.109***	0.113**	0.217***	0.244**
	(0.041)	(0.050)	(0.066)	(0.097)
Observations	7	7	7	7
Adj. R^2	0.418	0.352	0.443	0.457
Panel B: Provincial regressions				
	(1) MCB/GDP_{2012}	(2) MCB/GDP_{2013}	(3) MCB/GDP_{2014}	(4) MCB/GDP_{2015}
$BL/GDP^{stimulus}_{2009}$	0.107	0.132***	0.199***	0.149**
	(0.069)	(0.051)	(0.058)	(0.072)
Observations	30	30	30	30
Adj. R^2	0.103	0.315	0.338	0.168
Panel C: City-level regressions				
	(1) MCB/GDP_{2012}	(2) MCB/GDP_{2013}	(3) MCB/GDP_{2014}	(4) MCB/GDP_{2015}
$BL/GDP^{stimulus}_{2009}$	0.040**	0.036***	0.075***	0.056***
	(0.018)	(0.010)	(0.019)	(0.021)
Observations	325	325	325	325
Adj. R^2	0.068	0.070	0.137	0.073



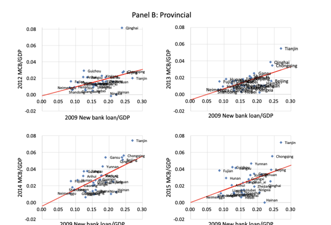


Fig. 4. 2009 stimulus new bank loan and 2012–2015 abnormal municipal corporate bond issuance. Panel A (B) presents the scatter plot with a fitted line for regional (provincial) data. Stimulus new bank loan and abnormal MCB issuance (as a percentage of GDP) are calculated over their average values between 2004 and 2008, respectively. The bank loan data are from the People's Bank of China, and the MCB-issuance data are from WIND.

What Fig. 4 adds. It visualizes the positive relationship in scatter plots (region-level and province-level). The pattern is consistent with the hangover hypothesis.

5.6 Table 4 and Fig. 5: main dynamic results (OLS vs IV)

Model link. These correspond to Eq. (4) with event-study coefficients β_τ.

How I read Table 4.

- *Pre-trends*: coefficients from 2004–2008 are close to zero (important for credibility).
- *After 2009*: coefficients become positive.
- *Post-2012*: coefficients become much larger and statistically significant, especially for total MCB issuance.

- *OLS vs IV*: IV magnitudes tend to exceed OLS magnitudes, consistent with a downward bias in OLS ("corrective endogeneity" logic: provinces with worse preconditions both receive fewer stimulus loans and later rely more on MCBs).

Purpose decomposition. The repayment component ($\text{MCB}^{\text{repay}}$) is strongly positive post-2012, consistent with the rollover channel. The investment component (MCB^{inv}) also increases, consistent with continuing-investment needs. **How I read Fig. 5.** It plots the $\hat{\beta}_\tau$ path with confidence intervals: no pretrend, delayed surge after 2012.

Table 4
Panel regressions: OLS and IV.
This table reports the results of provincial panel regressions of both OLS and IV. The dependent variables include MCB issuance, MCB issuance for bank loan repayment, MCB issuance for investment, and MCB issuance for other purposes, all of which are scaled by the 2009 GDP. $LateTerm_{2009}$ is used as the instrumental variable for the 2009 stimulus bank loan scaled by GDP. $LateTerm_{2009}$ equals one if a province governor was not in the first two years of his or her governor tenure as of 2009, and zero otherwise. Province fixed effects, year fixed effects, and the interaction terms of control variables and year dummies are included. Control variables include fiscal deficit scaled by GDP, fixed asset investment scaled by GDP, GDP growth, GDP per capita, and the Big Four branch share, the former four of which are measured over the one-year window of 2007Q4–2008Q3, and the last one is measured as of 2008Q3. Heteroskedasticity-consistent standard errors clustered by province and year are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The sample period is 2004–2015 without the benchmark year 2008.

	MCB/GDP_{2009}		MCB^{repay}/GDP_{2009}		MCB^{inv}/GDP_{2009}		MCB^{other}/GDP_{2009}	
	OLS (1)	2SLS (2)	OLS (3)	2SLS (4)	OLS (5)	2SLS (6)	OLS (7)	2SLS (8)
2004	−0.007	0.006	−0.001	−0.002	−0.007	0.005	0.001	0.002
	(0.016)	(0.045)	(0.005)	(0.013)	(0.008)	(0.022)	(0.002)	(0.005)
2005	−0.007	0.006	−0.001	−0.002	−0.006	0.005	0.001	0.003
	(0.016)	(0.046)	(0.005)	(0.013)	(0.009)	(0.022)	(0.002)	(0.005)
2006	−0.010	0.006	−0.001	−0.002	−0.009	0.005	0.001	0.002
	(0.015)	(0.046)	(0.005)	(0.013)	(0.008)	(0.023)	(0.002)	(0.005)
2007	−0.003	0.007	−0.001	−0.002	0.001	0.015	0.001	0.002
	(0.014)	(0.043)	(0.005)	(0.013)	(0.006)	(0.019)	(0.002)	(0.005)
2009	0.039***	0.103***	0.002	0.002	0.032***	0.087***	0.008***	0.018***
	(0.007)	(0.013)	(0.004)	(0.011)	(0.008)	(0.025)	(0.002)	(0.003)
2010	0.039**	0.139***	0.008	0.019***	0.025**	0.088***	0.009*	0.024***
	(0.017)	(0.029)	(0.005)	(0.007)	(0.012)	(0.034)	(0.004)	(0.007)
2011	0.080***	0.202***	0.016***	0.053***	0.041***	0.062**	0.010**	0.023*
	(0.018)	(0.039)	(0.003)	(0.010)	(0.008)	(0.024)	(0.004)	(0.012)
2012	0.091	0.144	0.023	0.016	0.026	0.122	0.010	0.022
	(0.058)	(0.161)	(0.022)	(0.049)	(0.043)	(0.108)	(0.011)	(0.023)
2013	0.275***	0.429**	0.100***	0.142	0.102***	0.176***	0.017**	0.022*
	(0.091)	(0.219)	(0.034)	(0.091)	(0.031)	(0.065)	(0.008)	(0.012)
2014	0.411***	0.784**	0.131***	0.231*	0.049	0.285***	0.057***	0.106**
	(0.140)	(0.358)	(0.051)	(0.133)	(0.050)	(0.109)	(0.015)	(0.053)
2015	0.411***	1.099***	0.195***	0.517***	0.002	0.158*	0.040*	0.165***
	(0.158)	(0.385)	(0.061)	(0.142)	(0.047)	(0.095)	(0.022)	(0.055)
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	360	360	360	360	360	360	360	360
$F^{effective}_{1st-stage}$		3.100		3.100		3.100		3.100
Adj. R^2	0.722	0.692	0.644	0.617	0.688	0.686	0.570	0.569

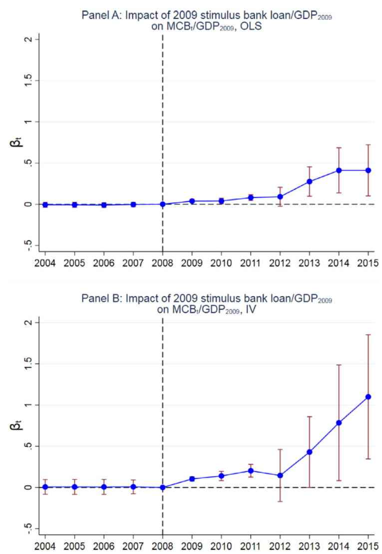


Fig. 5. Effects of 2009 stimulus bank loan on MCB issuance over 2009 GDP. Panels A and B plot the coefficients along with the 95% confidence intervals for the OLS and 2SLS regressions of MCB issuance over the 2009 GDP on the 2009 stimulus new bank loan scaled by GDP, respectively. The instrumental variable $LateTerm_{2009}$ equals one if a province governor was not in the first two years of his or her governor tenure as of 2009, and zero otherwise. Province fixed effects, year fixed effects, and the interaction terms of control variables and year dummies are included. Control variables include fiscal deficit scaled by GDP, fixed asset investment scaled by GDP, GDP growth, GDP per capita, and the Big Four branch share, the former four of which are measured over the one-year window of 2007Q4–2008Q3, and the last one is measured as of 2008Q3. Heteroskedasticity-consistent standard errors clustered by province and year are used to calculate the confidence intervals.

5.7 Table 5: city-level panel results (OLS vs IV)

What it adds. City-level evidence shows similar qualitative patterns. The paper also notes weak-IV concerns (low first-stage F), and discusses robustness via alternative weak-IV estimators.

5.8 Table 6: CDB-loan maturity heterogeneity

Model link. This is Eq. (5).

What it shows. Post-2012 effects are stronger in provinces with *low* CDB-loan fractions (i.e., fewer long-maturity policy-bank loans in 2009). This supports the rollover/maturity mechanism: shorter-maturity stimulus loans create more refinancing pressure a few years later.

5.9 Table 7: robustness with simultaneous controls

Idea. Add controls measured in the same year as MCB issuance (to address concerns that time-varying local conditions drive both stimulus-loan exposure and bond issuance).

Table 5
Panel regressions: OLS and IV at the city level.
This table reports the results of city-level panel regressions of both OLS and IV. The dependent variables include MCB issuance, MCB issuance for bank loan repayment, MCB issuance for investment, and MCB issuance for other purposes, all of which are scaled by the 2009 GDP. Both $LateTerm_{2009}$ and $LateTerm^{mayor}_{2009}$ are used as the instrumental variables for the 2009 stimulus bank loan scaled by GDP. $LateTerm_{2009}/LateTerm^{mayor}_{2009}$ equals one if a province governor/city mayor was not in the first two years of his or her tenure as of 2009, and zero otherwise. City fixed effects, year fixed effects, and the interaction terms of control variables and year dummies are included. Control variables include fiscal deficit scaled by GDP, fixed asset investment scaled by GDP, GDP growth, GDP per capita, the Big Four branch share, the former four of which are measured in 2008, and the last one is measured as of December 31, 2008. Heteroskedasticity-consistent standard errors clustered by city and year are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The sample period is 2004–2015 without the benchmark year 2008.

	MCB/GDP_{2009}		MCB^{repay}/GDP_{2009}		MCB^{inv}/GDP_{2009}		MCB^{other}/GDP_{2009}	
	OLS (1)	2SLS (2)	OLS (3)	2SLS (4)	OLS (5)	2SLS (6)	OLS (7)	2SLS (8)
2004	−0.002	0.000	0.000	0.000	−0.002	0.000	0.000	0.000
	(0.004)	(0.013)	(0.000)	(0.000)	(0.001)	(0.005)	(0.000)	(0.000)
2005	−0.001	0.001	0.000	0.000	−0.001	0.001	0.000	0.000
	(0.004)	(0.012)	(0.000)	(0.000)	(0.002)	(0.005)	(0.000)	(0.000)
2006	−0.001	0.001	0.000	0.000	−0.001	0.002	0.000	0.000
	(0.004)	(0.012)	(0.000)	(0.000)	(0.001)	(0.005)	(0.000)	(0.000)
2007	0.000	0.005	0.000	0.000	0.000	0.005	0.000	0.000
	(0.004)	(0.010)	(0.000)	(0.000)	(0.001)	(0.004)	(0.000)	(0.000)
2009	0.005	0.042***	0.000	0.001	0.004**	0.037***	0.000	0.002
	(0.003)	(0.005)	(0.001)	(0.003)	(0.002)	(0.008)	(0.000)	(0.001)
2010	0.009***	0.030***	0.001	0.004*	0.007***	0.013	0.002***	0.010***
	(0.003)	(0.003)	(0.001)	(0.002)	(0.002)	(0.012)	(0.001)	(0.001)
2011	0.016***	0.003	0.005***	0.012***	0.009*	−0.014	0.004***	0.005**
	(0.003)	(0.011)	(0.001)	(0.002)	(0.005)	(0.013)	(0.001)	(0.002)
2012	0.060**	0.074	0.019**	0.034***	0.026***	0.024	0.013***	0.014**
	(0.026)	(0.052)	(0.008)	(0.013)	(0.006)	(0.034)	(0.004)	(0.007)
2013	0.063***	0.100*	0.020***	0.050***	0.016***	0.007	0.008***	0.017***
	(0.017)	(0.056)	(0.008)	(0.014)	(0.006)	(0.029)	(0.003)	(0.006)
2014	0.138***	0.341***	0.059***	0.118***	0.026***	0.253***	0.020***	0.010
	(0.034)	(0.106)	(0.014)	(0.039)	(0.007)	(0.032)	(0.005)	(0.019)
2015	0.086**	0.393***	0.068***	0.208***	0.009**	0.089***	0.032***	0.099***
	(0.038)	(0.102)	(0.012)	(0.043)	(0.004)	(0.021)	(0.007)	(0.023)
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,900	3,648	3,900	3,648	3,900	3,648	3,900	3,648
$F^{effective}_{1st-stage}$		3.564		3.564		3.564		3.564
Adj. R^2	0.392	0.359	0.345	0.277	0.233	0.234	0.249	0.204

Table 6
Panel regressions: the effect of CDB loan fraction.
This table reports the results of provincial panel regressions for the interaction effect between 2009 stimulus bank loan and low CDB loan dummy. The dependent variables include MCB issuance, MCB issuance for bank loan repayment, MCB issuance for investment, and MCB issuance for other purposes, all of which are scaled by the 2009 GDP. $LateTerm_{2009}$ is used as the instrumental variable for the 2009 stimulus bank loan scaled by GDP. $LateTerm_{2009}$ equals one if a province governor was not in the first two years of his or her governor tenure as of 2009, and zero otherwise. The CDB^{low} is an indicator variable that takes the value of one if a province's 2009 CDB loan fraction is below the median value across all provinces. Province fixed effects, year fixed effects, and the interaction terms of control variables and year dummies are included. Control variables include fiscal deficit scaled by GDP, fixed asset investment scaled by GDP, GDP growth, GDP per capita, and the Big Four branch share, the former four of which are measured over the one-year window of 2007Q4–2008Q3, and the last one is measured as of 2008Q3. Heteroskedasticity-consistent standard errors clustered by province and year are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The sample period is 2004–2015 without the benchmark year 2008.

	MCB/GDP_{2009}		MCB^{repay}/GDP_{2009}		MCB^{inv}/GDP_{2009}		MCB^{other}/GDP_{2009}	
	OLS (1)	2SLS (2)	OLS (3)	2SLS (4)	OLS (5)	2SLS (6)	OLS (7)	2SLS (8)
2004	0.001	0.008	0.000	0.001	0.001	0.006	0.001	−0.001
	(0.017)	(0.059)	(0.007)	(0.018)	(0.010)	(0.030)	(0.003)	(0.007)
2005	0.001	0.011	0.000	0.001	0.001	0.008	0.000	−0.001
	(0.017)	(0.060)	(0.007)	(0.018)	(0.011)	(0.030)	(0.003)	(0.007)
2006	0.000	0.008	0.000	0.001	0.000	0.005	0.001	−0.001
	(0.017)	(0.059)	(0.007)	(0.018)	(0.010)	(0.030)	(0.003)	(0.007)
2007	0.005	0.007	0.000	0.001	0.008	0.010	0.001	−0.001
	(0.017)	(0.062)	(0.007)	(0.018)	(0.011)	(0.029)	(0.003)	(0.007)
2008	0.087***	0.123***	0.009	0.012	0.068***	0.103**	0.012***	0.018***
	(0.028)	(0.030)	(0.007)	(0.015)	(0.024)	(0.039)	(0.004)	(0.007)
2009	0.100***	0.213***	0.021***	0.030**	0.063**	0.158***	0.014***	0.029***
	(0.032)	(0.041)	(0.008)	(0.015)	(0.027)	(0.042)	(0.005)	(0.010)
2010	0.119***	0.208***	0.030***	0.064***	0.062**	0.074*	0.009	0.015
	(0.036)	(0.053)	(0.008)	(0.016)	(0.026)	(0.044)	(0.007)	(0.012)
2012	0.434***	0.276	0.109	−0.059	0.198***	0.113	0.040	0.038
	(0.152)	(0.321)	(0.090)	(0.143)	(0.065)	(0.117)	(0.034)	(0.059)
2013	0.661***	0.706*	0.200**	0.097	0.254***	0.187	0.048	0.046
	(0.165)	(0.385)	(0.090)	(0.142)	(0.068)	(0.127)	(0.031)	(0.056)
2014	0.880***	1.190***	0.245**	0.204	0.237***	0.224*	0.099***	0.196***
	(0.170)	(0.415)	(0.096)	(0.161)	(0.070)	(0.135)	(0.036)	(0.068)
2015	1.018***	1.479***	0.401***	0.558***	0.193***	0.121	0.115***	0.205***
	(0.169)	(0.418)	(0.093)	(0.148)	(0.067)	(0.140)	(0.036)	(0.067)
BL^{2009} × Subperiod	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
CDB^{low} × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control × Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	360	360	360	360	360	360	360	360
Adj. R^2	0.805	0.729	0.721	0.658	0.717	0.676	0.612	0.604

Result. Coefficients become smaller (possible over-controlling), but the post-2012 positive pattern remains.

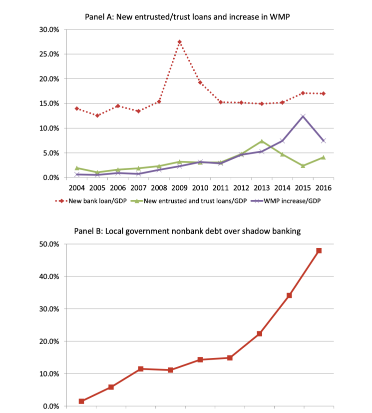


Fig. 6. Shadow banking activities and local government nonbank debt. Panel A plots new trust loans (entrusted and trust loans) and change in WMPs over GDP from 2004 to 2016. Aggregate new bank loans over GDP is also plotted for comparison. Panel B plots local government nonbank debt balance as a fraction of China's shadow banking balance from 2008 to 2016. Local government nonbank debt is the sum of MCBs, munibonds, and local government trust balance. Shadow banking balance is proxied by three items in the Aggregate Financing to the Real Economy, including trust loans (trust and entrusted loans), undiscounted bankers' acceptances, and corporate bonds. The annual WMP balance data are from the China Commercial Banks' Wealth Management Products Annual Report issued by the China Banking Wealth Management Registration System. The Aggregate Financing to the Real Economy by category data are from the PBoC.

Table 9
The effects of 2009 stimulus bank loan on future new entrust loans and bank loans.
This table reports the year-by-year regression results of new entrusted loans (EL) and abnormal bank loans (BL) on the 2009 stimulus bank loan. The dependent variable is the new entrusted loans scaled by GDP for Columns (1)–(4) and abnormal new bank loans scaled by GDP over its 2004–2008 average for Columns (5)–(8). The main explanatory variable is the 2009 stimulus bank loan scaled by GDP. Control variables include fiscal deficit scaled by GDP, fixed asset investment scaled by GDP, GDP growth, GDP per capita, and the Big Four branch share, the former four of which are measured over the one-year window of 2007Q4–2008Q3, and the last one is measured as of 2008Q3, or as of the same year as MCB issuance. Data on bank loans and entrusted loans are obtained from the PBoC, and data on control variables are obtained from the National Bureau of Statistics of China. Heteroskedasticity-consistent standard errors are reported in parentheses ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The sample period is 2013–2015.

	EL/GDP				$BL^{abnormal}$/GDP			
	OLS (1)	2SLS (2)	OLS (3)	2SLS (4)	OLS (5)	2SLS (6)	OLS (7)	2SLS (8)
2013	0.166**	0.270	0.228**	0.021	−0.083	−0.320	0.027	0.153
	(0.079)	(0.196)	(0.105)	(0.333)	(0.106)	(0.300)	(0.093)	(0.393)
$Control_{07Q4-08Q3}$	Yes	Yes	No	No	Yes	Yes	No	No
$Control_{simultaneous}$	No	No	Yes	Yes	No	No	Yes	Yes
Observations	30	30	30	30	30	30	30	30
Adj. R^2	0.400	0.365	0.306	0.157	0.549	0.513	0.565	0.555
2014	0.234***	0.124	0.279***	0.058	−0.118	−0.215	−0.014	0.022
	(0.079)	(0.331)	(0.093)	(0.269)	(0.124)	(0.331)	(0.091)	(0.300)
$Control_{07Q4-08Q3}$	Yes	Yes	No	No	Yes	Yes	No	No
$Control_{simultaneous}$	No	No	Yes	Yes	No	No	Yes	Yes
Observations	30	30	30	30	30	30	30	30
Adj. R^2	0.3432	0.309	0.290	0.138	0.602	0.598	0.547	0.547
2015	−0.025	0.113	0.082	0.096	−0.203	−0.727	−0.024	0.039
	(0.093)	(0.208)	(0.114)	(0.224)	(0.166)	(0.519)	(0.184)	(0.354)
$Control_{07Q4-08Q3}$	Yes	Yes	No	No	Yes	Yes	No	No
$Control_{simultaneous}$	No	No	Yes	Yes	No	No	Yes	Yes
Observations	30	30	30	30	30	30	30	30
Adj. R^2	0.324	0.261	0.098	0.097	0.416	0.321	0.427	0.426

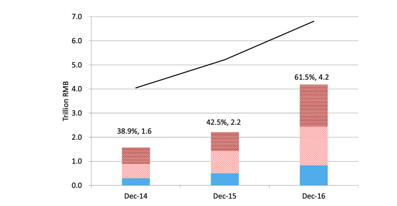


Fig. 7. Wealth management products investment in municipal corporate bonds, from 2014 to 2016. The solid line plots the total MCB balance, and bars represents WMPs invested in MCBs with various ratings. The percentage and RMB value (in trillions) of MCBs held by WMPs are given above the bars. The MCB data are from WIND, and the WMP investment data are from the China Commercial Banks' Wealth Management Products Annual Report issued by the China Banking Wealth Management Registration System. More estimation details are in Online Appendix D.

5.10 Table 8: real outcomes (GDP per capita and investment)

What it tries to do. Replace the dependent variable in Eq. (4) with GDP per capita (scaled by 2009) and fixed asset investment (scaled by 2009 GDP).

How I interpret it.

- There is some evidence that stimulus loans correlate with later GDP per capita and investment.
- Because the IV is weak, I treat large IV magnitudes as suggestive rather than decisive.

5.11 Fig. 6, Table 9, Fig. 7: linking stimulus exposure to shadow banking

Fig. 6. Shadow-banking aggregates (trust/entrusted loans and WMP) grow strongly after 2012, and local-government nonbank debt becomes a larger share of shadow banking.

Table 9. Provinces with larger 2009 stimulus-loan shocks later exhibit higher entrusted-loan growth (a shadow-banking channel), while the relationship with abnormal bank-loan growth is weak/negative. This fits the substitution story: financing demand migrates away from standard bank balance sheets. **Fig. 7.** WMPinvestment in MCBs rises sharply from 2014 to 2016, reaching

Table 7

Panel regressions: OLS and IV with simultaneous controls.

This table reports the results of provincial panel regressions of both OLS and IV. The dependent variables include MCB issuance, MCB issuance for bank loan repayment, MCB issuance for investment, and MCB issuance for other purposes, all of which are scaled by the 2009 GDP. $LateTerm_{2009}$ is used as the instrumental variable for the 2009 stimulus bank loan scaled by GDP. $LateTerm_{2009}$ equals one if a province governor was not in the first two years of his or her governor tenure as of 2009, and zero otherwise. Province fixed effects, year fixed effects, and simultaneous control variables are included. Control variables include fiscal deficit scaled by GDP, fixed asset investment scaled by GDP, GDP growth, GDP per capita, and the Big Four branch share, all of which are as of the same year as MCB issuance. Heteroskedasticity-consistent standard errors clustered by province and year are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The sample period is 2004–2015 without the benchmark year 2008.

	MCB/GDP_{2009}		MCB^{repay}/GDP_{2009}		MCB^{inv}/GDP_{2009}		MCB^{other}/GDP_{2009}	
	OLS (1)	2SLS (2)	OLS (3)	2SLS (4)	OLS (5)	2SLS (6)	OLS (7)	2SLS (8)
2004	−0.009	−0.012	0.013	0.020	−0.022*	−0.033	0.006	0.010
	(0.030)	(0.062)	(0.012)	(0.021)	(0.011)	(0.020)	(0.003)	(0.008)
2005	0.022	0.020	0.019*	0.023	−0.007	−0.014	0.009***	0.013*
	(0.022)	(0.050)	(0.010)	(0.017)	(0.008)	(0.018)	(0.002)	(0.007)
2006	−0.003	−0.002	0.012	0.015	−0.019	−0.024*	0.004	0.007
	(0.027)	(0.039)	(0.011)	(0.013)	(0.011)	(0.013)	(0.003)	(0.005)
2007	−0.004	0.008	0.005	0.010	−0.009	−0.008	0.001	0.004
	(0.027)	(0.044)	(0.009)	(0.012)	(0.011)	(0.019)	(0.005)	(0.008)
2009	−0.006	−0.017	−0.008	−0.013	0.001	−0.012	0.005	0.004
	(0.025)	(0.057)	(0.010)	(0.020)	(0.014)	(0.026)	(0.003)	(0.006)
2010	−0.034	−0.035	−0.018	−0.027	−0.008	−0.006	−0.001	0.001
	(0.051)	(0.099)	(0.023)	(0.044)	(0.019)	(0.039)	(0.006)	(0.010)
2011	0.017	0.015	−0.004	−0.007	0.005	−0.012	0.001	0.001
	(0.062)	(0.125)	(0.030)	(0.059)	(0.018)	(0.040)	(0.007)	(0.014)
2012	0.116	0.207	0.032	0.061	−0.011	−0.026	0.025*	0.057**
	(0.102)	(0.185)	(0.028)	(0.061)	(0.034)	(0.054)	(0.013)	(0.027)
2013	0.189**	0.149	0.090***	0.101*	0.048**	0.008	0.020*	0.032
	(0.076)	(0.118)	(0.031)	(0.061)	(0.021)	(0.035)	(0.011)	(0.021)
2014	0.362***	0.380**	0.143***	0.185**	0.050*	0.106**	0.052***	0.057
	(0.097)	(0.157)	(0.040)	(0.077)	(0.030)	(0.054)	(0.019)	(0.040)
2015	0.255**	0.185	0.135**	0.117*	−0.011	0.013	0.036*	0.056
	(0.128)	(0.180)	(0.054)	(0.067)	(0.029)	(0.039)	(0.020)	(0.039)
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	360	360	360	360	360	360	360	360
Adj. R^2	0.743	0.711	0.659	0.625	0.674	0.674	0.555	0.546

Table 8

Panel regressions: real effects.

This table reports the results of provincial panel regressions of both OLS and IV. The dependent variables include GDP per capita scaled by the 2009 GDP per capita and fixed asset investment scaled by the 2009 GDP. $LateTerm_{2009}$ is used as the instrumental variable for the 2009 stimulus bank loan scaled by GDP. $LateTerm_{2009}$ equals one if a province governor was not in the first two years of his or her governor tenure as of 2009, and zero otherwise. Province fixed effects, year fixed effects, and the interaction terms of control variables and year dummies are included. Control variables include fiscal deficit scaled by GDP, fixed asset investment scaled by GDP, GDP growth, GDP per capita, and the Big Four branch share, the former four of which are measured over the one-year window of 2007Q4–2008Q3, and the last one is measured as of 2008Q3. Heteroskedasticity-consistent standard errors clustered by province and year are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The sample period is 2004–2015 without the benchmark year 2008.

	$GDP^{capita}/GDP^{capita}_{2009}$		FAI/GDP_{2009}	
	OLS (1)	2SLS (2)	OLS (3)	2SLS (4)
2004	0.037	0.799***	0.142	0.720*
	(0.156)	(0.245)	(0.242)	(0.399)
2005	−0.057	0.306	0.059	0.419
	(0.153)	(0.265)	(0.247)	(0.429)
2006	−0.107	0.103	0.012	0.207
	(0.164)	(0.280)	(0.203)	(0.390)
2007	−0.012	0.369	−0.047	0.142
	(0.102)	(0.247)	(0.156)	(0.355)
2009	−0.108**	0.466***	0.460*	−0.084
	(0.055)	(0.128)	(0.259)	(0.378)
2010	0.031	0.486**	0.704**	0.257
	(0.082)	(0.194)	(0.245)	(0.490)
2011	0.259*	0.815*	0.708***	−0.259
	(0.137)	(0.435)	(0.274)	(0.849)
2012	0.321	1.647**	0.645*	−0.251
	(0.265)	(0.792)	(0.366)	(1.096)
2013	0.322	2.808**	0.729	−0.836
	(0.411)	(1.358)	(0.519)	(1.456)
2014	0.312	4.098**	0.452	−0.600
	(0.614)	(1.936)	(0.696)	(1.714)
2015	0.372	5.584**	1.405*	4.156**
	(0.721)	(2.202)	(0.831)	(1.844)
Province FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Control × Year	Yes	Yes	Yes	Yes
Observations	360	360	360	360
Adj. R^2	0.978	0.981	0.971	0.971

a large fraction of the MCB market. This is a concrete "plumbing" link: WMPs are a funding source that absorbs local-government bond issuance.

5.12 Fig. 8: bond-market development

What I see. The number of credit-rating reports and sell-side bond-market research reports rises dramatically after 2012, and the interbank bond market sees rising participation by nonbank financial institutions.

Why it matters. The hangover effect is interpreted as a catalyst for the expansion of market infrastructure and nonbank participation in bond markets.

6 Short summary

- The 2009 stimulus generated a large stock of LGFV-related bank loans that created delayed refinancing needs.
- When bank credit tightens later, these needs manifest as rapid growth of MCB issuance and shadow banking after 2012.
- The paper does *not* provide a full structural general-equilibrium quantification of shadow banking's welfare impact, but it provides causal evidence consistent with the "hangover" mechanism and documents how it coincides with bond-market development.

7 Conclusion

7.1 Summary

This paper documents and explains a delayed transition in the financing of Chinese local governments. The central empirical claim is that China's 2009 credit stimulus created a large stock of bank loans—especially LGFV-related loans—that generated *future refinancing and continuation needs*. When the traditional bank-loan channel tightened later (especially after 2012), these financing needs *migrated* into (i) **municipal corporate bonds (MCBs)** issued by LGFVs and (ii) **shadow-banking channels** (entrusted loans, trust-related credit, and wealth-management products, WMPs). Hence the paper interprets the post-2012 rise of bond-financed quasi-municipal debt and shadow banking as a **"stimulus-loan hangover": the stimulus loans wane, but the refinancing demand persists and reappears elsewhere in the financial system.**

7.2 Key empirical conclusions (with quantitative anchors)

1. **Local-government debt composition shifts away from bank loans toward nonbank debt.** From the NAO audits (Table 1), total local government debt rises from RMB 10,717 bn (2010/12/31) to RMB 17,891 bn (2013/6/30), while the *bank-loan share* falls from 79.0% to 56.6%. New nonbank categories become large (e.g., trust reaches 8.0%, build-and-transfer 8.3%, and "others" 10.5%), and bond share rises from 7.1% to 10.3%.
2. **Stimulus magnitude is economically large.** In the paper's accounting, there are about RMB 4.7 trillion "extra" new bank loans in 2009, of which LGFVs receive roughly RMB 2.3 trillion (about RMB 2.06 trillion from commercial banks and RMB 0.26 trillion from policy banks).
3. **Cross-sectional evidence: provinces with larger 2009 stimulus lending later issue more MCBs (2012–2015).** In year-by-year regressions (Table 3), the coefficient linking

stimulus lending intensity to abnormal MCB issuance is positive and becomes stronger post-2012. At the provincial level, the slope estimates (2012–2015) are approximately:

$$\hat{b}_{2012} \approx 0.107, \quad \hat{b}_{2013} \approx 0.132^{***}, \quad \hat{b}_{2014} \approx 0.199^{***}, \quad \hat{b}_{2015} \approx 0.149^{**},$$

with similar positive patterns at the region and city levels.

4. **Dynamic panel evidence: no pre-trends, then a delayed post-2012 surge.** In the event-study panel (Table 4; Fig. 5), coefficients in 2004–2007 are close to zero, supporting the absence of pre-trends. Effects are modest in 2009–2011 and become large after 2012. For total MCB issuance (scaled by 2009 GDP), the OLS coefficients for the post-2012 period are large, e.g.

$$\hat{\beta}_{2013} \approx 0.275^{***}, \quad \hat{\beta}_{2014} \approx 0.411^{***}, \quad \hat{\beta}_{2015} \approx 0.411^{***}.$$

IV magnitudes are even larger (e.g. $\hat{\beta}^{IV}_{2015} \approx 1.099^{***}$), but the paper cautions that the IV first stage is weak (effective first-stage $F \approx 3.1$), so IV magnitudes should be interpreted carefully.

5. **Purpose decomposition supports the "rollover + continuation" mechanism.** The post-2012 rise in MCB issuance is materially explained by bonds used for *bank-loan repayment/rollover* and (to a lesser extent) investment. In Table 4, the repayment component has strong post-2012 coefficients (e.g. OLS $\hat{\beta}^{repay}_{2013} \approx 0.100^{***}$, $\hat{\beta}^{repay}_{2014} \approx 0.131^{***}$, $\hat{\beta}^{repay}_{2015} \approx 0.195^{***}$).

6. **Maturity/rollover channel: the effect is stronger when 2009 lending is shorter-maturity.** The heterogeneity test using the fraction of 2009 policy-bank (CDB) loans (Table 6) shows that the post-2012 effect is stronger in provinces with *low* CDB-loan shares (i.e., more reliance on shorter-maturity commercial-bank lending), consistent with refinancing pressure being a key driver rather than a generic "stimulus demand" story.

7. **The refinancing pressure spills into shadow banking rather than returning to standard bank lending.** In Table 9, the 2009 stimulus lending shock predicts *new entrusted-loan growth* in 2013–2014:

$$\widehat{\gamma}_{2013} \approx 0.166^{**}, \qquad \widehat{\gamma}_{2014} \approx 0.234^{***} \quad \text{(EL/GDP, with baseline controls)},$$

while the relationship with abnormal new bank-loan growth is weak/negative in sign. This pattern supports a substitution/migration story: financing demand shifts from on-balance-sheet bank loans to nonbank channels.

8. **WMPs become a major funding base for MCBs (a plumbing link).** Figure 7 shows WMP holdings of MCBs rise from about 38.9% (RMB 1.6 tn) in Dec 2014 to 42.5% (RMB 2.2 tn) in Dec 2015, and to 61.5% (RMB 4.2 tn) in Dec 2016, highlighting how shadow-banking liabilities fund quasi-municipal bonds.

7.3 Economic intuition: the "stimulus-loan hangover" mechanism

The paper's results are coherent under a simple two-step logic:

- **Step 1 (creation of future refinancing needs).** The 2009 stimulus produces a large, uneven stock of LGFV-related bank loans. Even if the stimulus initially finances real investment, these loans have maturities and require rollover/repayment several years later.
- **Step 2 (migration when bank lending tightens).** As bank credit conditions tighten (and as policy restricts new LGFV borrowing for new projects), LGFVs and local governments increasingly use MCB issuance and shadow-banking instruments to meet (i) rollover needs and (ii) continuation financing for multi-year projects. The delayed timing (post-2012) is therefore a *maturity/rollover* phenomenon rather than an immediate contemporaneous shock.

7.4 Main implication

A large, policy-driven credit expansion can have **long-lived aftereffects**: even when the direct policy stimulus fades and traditional bank lending retrenches, the financing pressure remains and reappears in new forms (bonds + shadow banking), potentially changing the structure of the financial system and the locus of risk.

7.5 Takeaway

China's 2009 stimulus lending created a delayed refinancing "hangover" that, once bank lending tightened, shifted local-government financing toward LGFV bond issuance and shadow banking after 2012, with WMPs becoming a major buyer base for quasi-municipal bonds.